

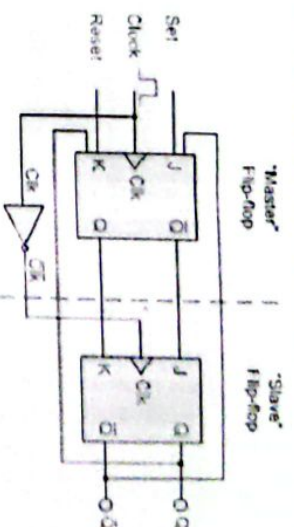
UNIT III

1. Distinguish between synchronous and asynchronous sequential circuits. (APRIL 2011)

SYNCHRONOUS CIRCUIT	SEQUENTIAL	ASYNCHRONOUS CIRCUIT	SEQUENTIAL
state variables : F/Fs □ controlled by a clock □ operate in pulse mode		do not operate in pulse mode □ do not use F/Fs to represent state variables □ Changes in state are dependent on whether each of inputs to the circuit has the logic level 0 or 1 at any given time	
BLOCK DIAGRAM		BLOCK DIAGRAM	
			
Clock pulse is applied simultaneously.		Clock pulse is applied to the first flip flop; the change of output is given as clock to next flip flop.	
Speed of operation is high		Speed of operation is low.	

2. Why J-K Flip Flop is called Master Flip Flop? (APRIL 2011)

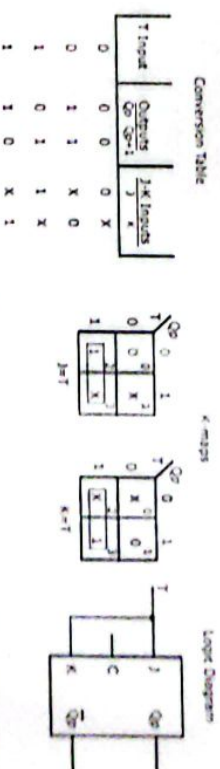
Master-slave flip flop is designed using two separate flip flops. Out of these, one acts as the master and the other as a slave. The figure of a master-slave J-K flip flop is shown below. From the above figure you can see that both the J-K flip flops are presented in a series connection. The output of the master J-K flip flop is fed to the input of the slave J-K flip flop. The output of the slave J-K flip flop is given as a feedback to the input of the master J-K flip flop. The clock pulse [Ck] is given to the master J-K flip flop and it is sent through a NOT gate and thus inverted before passing it to the slave J-K flip flop.



3. How can you obtain T Flip-Flop from JK Flip-Flop and draw logic diagram? (NOV 2011)

J and K are the actual inputs of the flip flop and T is taken as the external input for conversion. Four combinations are produced with T and Qp. J and K are expressed in terms of T and Qp. The conversion table, K-maps, and the logic diagram are given below.

J-K Flip Flop to T Flip Flop



4. What is Shift Register? (NOV 2011)

A register capable of shifting the binary information held in each cell to its neighboring cell, in a selected direction, is called a *shift register*. The logical configuration of a shift register consists of a chain of flip-flops in cascade, with the output of one flip-flop connected to the input of the next flip-flop. All flip-flops receive common clock pulses, which activate the shift of data from one stage to the next.

5. Draw the logic diagram of SR flip-flop. (NOV 2012)



6. Define Flip-Flop and write some common types of Flip-Flops. (APRIL/MAY 2012)

What is Flip-Flop? (APRIL 2013 - 11)

The memory elements used in clocked sequential circuits are called *flip-flops*. These circuits are binary cells capable of storing one bit of information. Binary information can enter a flip-flop in a variety of ways a few that gives rise to different types of flip-flops.

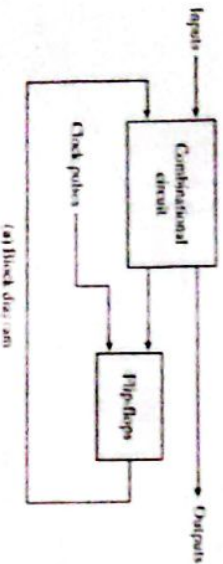
- > D flip flop
- > T flip flop
- > JK flip flop
- > SR flip flop

7. Explain Binary Ripple counters. (APRIL/MAY 2012)

A binary ripple counter consists of a series connection of complementing flip-flops, with the output of each flip-flop connected to the $\bar{C}$ input of the next higher order flip-flop. The flip-flop holding the least significant bit receives the incoming count pulses.

8. What is meant by clocked sequential circuits? (APRIL 2013)

Synchronous sequential circuits that use clock pulses to control storage elements are called *clocked sequential circuits* and are the type most frequently encountered in practice. They are called *synchronous circuits* because the activity within the circuit and the resulting updating of stored values is synchronized to the occurrence of clock pulses



9. Define the term counter. (APRIL 2013)

A register that goes through a prescribed sequence of states upon the application of input pulses is called a *counter*. The input pulses may be clock pulses, or they may originate from some external source and may occur at a fixed interval of time or at random. An n -bit binary counter consists of n flip-flops and can count in binary from 0 through $2^n - 1$. Counters are available in two categories, ripple counters and synchronous counters.

10. What is Latches? (APRIL 2013 - 11)

A latch is an electronic logic circuit that has two inputs and one output. One of the inputs is called the SET input, the other is called the RESET input.

Latch circuits can be either active-high or active-low. The difference is determined by whether the operation of the latch circuit is triggered by HIGH or LOW signals on the inputs.

11. What is edge-triggered flip-flop? (APRIL 11)

The problem of race around condition can be solved by edge triggering flip flop. The term edge triggering means that the flip-flop changes state either at the positive edge or negative edge of the clock pulse and it is sensitive to its inputs only at this transition of the clock.

The three basic types are introduced here: S-R, J-K and D.

12. What are hazards? (or) Define essential hazard. (NOV 2011) (APRIL/MAY 2014).

Hazards in any system are obviously an undesirable effect caused by either a deficiency in the system or external influences. In digital logic hazards are usually referred to in one of three ways:

- Static Hazards
- Dynamic Hazards
- Function Hazards

These logic hazards are all subsets of the same problem: - When changes in the input variables do not change the output due to some form of delay caused by logic elements (NOT, AND, OR gates etc), this results in the logic not performing its function properly.

13. What is non critical race?

- > final stable state does not depend on the order in which the state variable changes
- > race condition is not harmful

14. What is critical race?

- > final stable state depends on the order in which the state variable changes
- > race condition is harmful

15. When does a cycle occur?

- > asynchronous circuit makes a transition through a series of unstable state

16. What are the different techniques used in state assignment?

- > shared row state assignment
- > one hot state assignment

17. What are the types of asynchronous circuits?

Types of asynchronous circuits

Fundamental mode circuits, Pulse mode circuits

18. What is static 1 hazard?

Output goes momentarily 0 when it should remain at 1

19. What is static 0 hazard?

Output goes momentarily 1 when it should remain at 0

20. What is dynamic hazard?

Output changes 3 or more times when it changes from 1 to 0 or 0 to 1

21. What is the cause for essential hazards?

Unequal delays along 2 or more path from same input.

22. Define race around condition and explain how it can be eliminated. (APRIL/MAY 2014).

A *Race condition* is when a device's output depends on two (or more) nearly simultaneous events to occur, and where which signal arrives first will change the output of the circuit. If the race condition is so close in time, the output may be unpredictable. When the circuit is manufactured, slight differences can cause a change in the operation of the circuit. A race condition can be a logic hazard, or can result in a random value being held in a latch.

Elimination of race condition

- > Races may be avoided by making a proper binary assignment to the state variables. The state variables must be assigned binary numbers in such a way that only one state variable can change at any one time when a state transition occurs in the flow table.
- > Races can be avoided by directing the circuit through intermediate unstable states with a unique state-variable change. When a circuit goes through a unique sequence of unstable states, it is said to have a cycle.

23. What is register and state its usage. (NOV 2011)

A *register* is a group of flip-flops, each one of which shares a common clock and is capable of storing one bit of information. An *n*-bit register consists of a group of *n* flip-flops capable of storing *n* bits of binary information. In addition to the flip-flops, a register may have combinational gates that perform certain data-processing tasks. In its broadest definition, a register consists of a group of flip-flops together with gates that affect their operation.

24. Define Blocking assignment statement. (APRIL, 2013 - 11)

Blocking assignment statements are executed *sequentially* in the order that they are listed in a sequential block; when they execute, they have an immediate effect on the contents of memory.

before the next statement can be executed. A general rule is to use blocking assignments when sequential ordering is imperative and in cyclic behavior that is level sensitive (i.e., incombinational logic).

25. Define combinational circuit.

A combinational circuit consist of logic gates whose output are at anytime are determined directly from the present combination of inputs without regard to previous inputs.

26. Define sequential circuits.

The outputs are a function of the inputs and the state of memory elements. The state of memory elements, in turn, is a function of previous inputs.

27. What do you mean by triggering of flip flop.

The state of a flip flop is switched by a momentary change in the input signal. This momentary change is called a trigger and the transition it causes is said to trigger the flip flop.

28. What is an excitation table?

During the design process we usually know the transition from present state to next state and wish to find the flip flop input conditions that will cause the required transition. A table which lists the required inputs for a given change of state is called an excitation table.

29. Give the excitation table of a JK and SR flip flop.

Q(0)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK flip - flop

Q(0)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	0	0

SR flip - flop

30. Give the excitation table of a T and D flip flop

Q(0)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

T flip flop

Q(0)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

D flip flop

31. What is a characteristic table? Give the characteristic equation of a SR flip flop, D flip flop, JK flip flop & T flip flop

A characteristic table defines the logical property of the flip flop and completely characterise its operation.

Characteristic equation

$$\text{SR flip flop } Q(t+1) = S + R \cdot Q$$

$$\text{D flip flop } Q(t+1) = D$$

$$\text{JK flip flop } Q(t+1) = JQ + K'Q$$

$$\text{T flip flop } Q(t+1) = TQ + T'Q'$$

32. What is the difference between truth table and excitation table.

- An excitation table is a table that lists the required for a given change of state.
- A truth table is a table that including the output of a logic circuit for various input state.

33. What is synchronous counter?

In a synchronous counter, the clock pulse is applied simultaneously to all flip flops. The output of the flip flop change state at the same instant. The speed of operation is high compared to an asynchronous counter.

34. What is asynchronous counter?

In an asynchronous counter, the clock pulse is applied to the first flip flop. The change of state in the output of this flip flop serves as a clock pulse to the next flip flop and so on. Here all the flip flops do not change state at the same instant and hence speed is less.

35. Name the different types of counter.

- Synchronous counter
- Asynchronous counter
 - UP counter
 - DOWN counter
 - Modulo - N counter
 - UP/DOWN counter

36. What is up counter?

A counter that increments the output by one binary number each time a clock pulse is applied.

37. What is down counter?

- A counter that decrements the output by one binary number each time a clock pulse is applied.

38. What is up/down counter?

- A counter, which is capable of operating as an up counter or down counter, depending on a control lead.

39. Form the truth table for 3-bit binary down counter.

clk	Q2	Q1	Q0
1	1	1	1
1	1	1	0
1	1	0	1
1	1	0	0
1	0	1	1
1	0	1	0
1	0	0	1
1	0	0	0
1	1	1	1

40. What is a ring counter? What are the uses of a ring counter?

A counter formed by circulating a 'bit' in a shift register whose serial output has been connected to its serial input.

- Uses of a ring counter
 - Control section of a digital system.
 - Controlling events, which occur in strict time sequence.

41. What is BCD counter?

A BCD counter counts in binary coded decimal from 0000 to 1001 and back does to 0000. Because of the return to 0000 after a count of 1001, from a BCD counter does not have a regular pattern as in a straight binary counter.

42. What is Johnson counter?

It is a ring counter in which the inverted output is fed into the input. It is also known as a twisted ring counter.

43. Write the uses of a shift register.

- Temporary data storage

- Bit manipulations.

44. What are the problems involved in asynchronous circuits?

- Cycles
- Races
- Hazard

45. Define Mealy & Moore model

In the Mealy model, the output is a function of both the present state and the input. In the Moore model, the output is a function of only the present state. A circuit may have both types of outputs. The two models of a sequential circuit are commonly referred to as a finite state machine, abbreviated FSM. The Mealy model of a sequential circuit is referred to as a Mealy FSM or Mealy machine. The Moore model is referred to as a Moore FSM or Moore machine.

46. What are the types of Sequential Circuits?

There are two basic types of sequential circuits that account for the majority of practical discrete designs:

1. feedback sequential circuit: uses ordinary gates and feedback loops to create sequential circuit building blocks such as latches and flip flops
2. clocked synchronous state machine: uses latches and flip flops (in particular, edge triggered "D" flip flops) to create circuits whose inputs are examined and whose outputs change state in accordance with a controlling clock signal.

47. List the application of shift register.

- One application of shift registers is converting between "serial data" and "parallel data."
- Recent switch from Parallel ATA to Serial ATA in hard drives.

48. What are the types of shift registers?

- Serial out shift register
- Parallel in parallel out shift register

UNIT - III

Sequential Logic: Latches versus Flip Flops – SR, D, JK, Master Slave Flip Flops – Excitation table – Conversion of Flip flops – Counters: Asynchronous, synchronous, decade, presettable – Shift Registers: types, applications – Ring counter – Analysis and design of clocked sequential circuits – Mealy and Moore models – State machine notations – state reduction techniques.

II MARKS

1. Explain in detail about the SR latch-SR Flip. (11) (APR13 - IT)

An SR latch is shown in figure 13.3. The latch Truth table is shown in the following table. The two inputs, S and R denote "set" and "reset" respectively.

The latch has memory, and the present output is dependent on the state of the latch. Thus the output at n^{th} instant, denoted by Q_n is dependent on output at $(n-1)^{th}$ instant, denoted by Q_{n-1} .

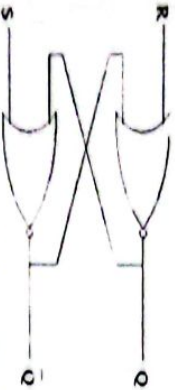


Figure 13.3: Construction of a latch from NOR gates

S	R	Q_n	$\overline{Q}_n$
1	0	1	0
0	1	0	1
1	1	0	0
0	0	Q_{n-1}	$\overline{Q}_{n-1}$

Note that in $SR = 11$ state, Q_n both $\overline{Q}_n$ and are 0, which seems absurd. Thus, conventionally, the state $SR = 11$ is said to be "not allowed".

A similar latch known as $\overline{S}\overline{R}$ latch is constructed using NAND gates (as opposed to NOR gates for SR latch). The students should again check that the working of the latch conforms with that of the truth table.



Figure 13.4: Construction of a latch from NAND gates

$\overline{S}$	$\overline{R}$	Q_n	$\overline{Q}_n$
0	1	1	0
1	0	0	1
0	0	1	1
1	1	Q_{n-1}	$\overline{Q}_{n-1}$

To avoid "race" between the inputs, to have a control on when the input affects the latch, the circuit 13.5 is often implemented.

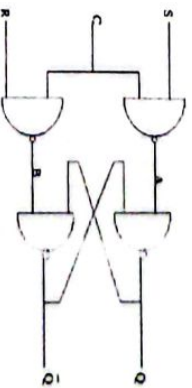


Fig. SR flip flop

S	R	Q_{n+1}
0	0	Q_n
0	1	1
1	0	0
1	1	*

Q_n	Q_{n+1}	R	S
0	0	X	0
0	1	0	1
1	0	1	0
1	1	0	X

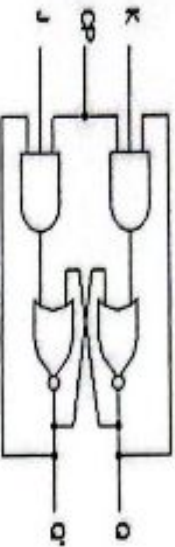
Table(a): RS truth table

Table(b): RS Excitation table

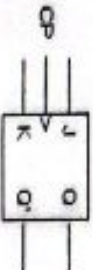
The inputs have an effect on the latch only when $C = 1$, otherwise, the previous state is maintained. The input C may be a clock, so that whatever transitions in S and R take place before the clock C changes to 1 do not affect the outputs, and only when the inputs have become stable is the system affected.

2. Describe the design procedure of JK flip-flop (5) (APR/MAY'12)

The circuit diagram and truth-table of a J-K flip flop is shown below.



(a) Logic diagram



(b) Graphical symbol

Q	J	K	Qn+1
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(c) Transition table

Clocked JK flip-flop

- > The behavior of inputs J and K is same as the S and R inputs of the S-R flip flop.
- > The letter J stands for SET and the letter K stands for CLEAR.
- > When both the inputs J and K have a HIGH state, the flip-flop switch to the complement state.
- > So, for a value of $Q = 1$, it switches to $Q = 0$ and for a value of $Q = 0$, it switches to $Q = 1$.
- > The circuit includes two 3-input AND gates.
- > The output Q of the flip flop is returned back as a feedback to the input of the AND along with other inputs like K and clock pulse [CP].

- > So, if the value of CP is '1', the flip flop gets a CLEAR signal and with the condition the value of Q was earlier 1.
- > Similarly output Q' of the flip flop is given as a feedback to the input of the AND along with other inputs like J and clock pulse [CP].
- > So the output becomes SET when the value of CP is 1 only if the value of Q' was earlier 1.
- > The output may be repeated in transitions once they have been complemented for J-K=1 because of the feedback connection in the JK flip-flop.
- > This can be avoided by setting a time duration lesser than the propagation delay through the flip-flop.
- > The restriction on the pulse width can be eliminated with a master-slave or edge-triggered construction.

J	K	Qn	Qn+1
0	0	0	0
0	1	0	0
1	0	1	1
1	1	1	0

TRUTH TABLE

Qn	Qn+1	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

EXCITATION TABLE

3. What is a Flip-Flop? Explain the working principle of D Flip-Flop, T Flip-Flop in detail. (11) (APR'12)

D FLIP FLOP

The circuit diagram and truth table is given below.



(a) Logic diagram with NAND gates



(b) Graphical symbol

Q	D	Qn+1
0	0	0
0	1	1
1	0	0
1	1	1

(c) Transition table

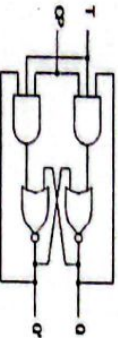
Clocked D Flip-Flop

- > D flip flop is actually a slight modification of the above explained clocked SR flip flop.

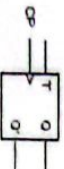
- > From the figure you can see that the D input is connected to the S input and the complement of the D input is connected to the R input.
- > The D input is passed on to the flip flop when the value of CP is '1'.
- > When CP is HIGH, the flip flop moves to the SET state. If it is '0', the flip flop switches to the CLEAR state.

T FLIP FLOP

- > This is a much simpler version of the J-K flip flop. Both the J and K inputs are connected together and thus are also called a single input J-K flip flop.
- > When clock pulse is given to the flip flop, the output begins to toggle.
- > Here also the restriction on the pulse width can be eliminated with a master-slave or edge-triggered construction.
- > Take a look at the circuit and truth table below.



(a) Logic diagram



(b) Graphical symbol

Q	T	Qn+1
0	0	0
0	1	1
1	0	0
1	1	1

(c) Transition table

Clocked T flip-flop

4. Explain in detail about analysis of clocked sequential circuits.

- > Analysis describes what a given circuit will do under certain operating conditions.
- > The behavior of a clocked sequential circuit is determined from the inputs, the outputs, and the state of its flip-flops.
- > The outputs and the next state are both a function of the inputs and the present state.
- > The analysis of a sequential circuit consists of obtaining a table or a diagram for the time sequence of inputs, outputs, and internal states.
- > It is also possible to write Boolean expressions that describe the behavior of the sequential circuit. These expressions must include the necessary time sequence, either directly or indirectly.

- > A state table and state diagram are then presented to describe the behavior of the sequential circuit.

The procedure for analysis of synchronous sequential circuits can be summarized by a list of recommended steps:

1. From the given logic diagram, obtain the state equation.
2. From the state equation, generate the state table.
3. Draw the state diagram using state table.
4. The steps presented in this example are summarized below:

Circuit diagram- State Equations - State table - State diagram

State equation

- > The behavior of a clocked sequential circuit can be described algebraically by means of state equations.
- > A *state equation* (also called a *transition equation*) specifies the next state as a function of the present state and inputs.

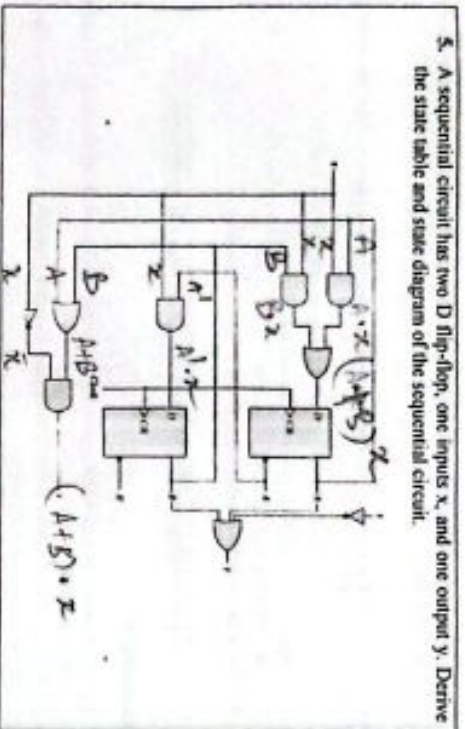
State Table

- > The time sequence of inputs, outputs, and flip-flop states can be enumerated in a *state table* (sometimes called a *transition table*).
- > The table consists of four sections labeled *present state*, *input*, *next state*, and *output*.
- > The present-state section shows the states of flip-flops *A* and *B* at any given time *t*.
- > The input section gives a value of *x* for each possible present state.
- > The next-state section shows the states of the flip-flops one clock cycle later, at time *t + 1*.
- > The output section gives the value of *y* at time *t* for each present state and input condition.

State Diagram

- > The information available in a state table can be represented graphically in the form of a state diagram.
- > In this type of diagram, a state is represented by a circle, and the (clock-triggered) transitions between states are indicated by directed lines connecting the circles.
- > The binary number inside each circle identifies the state of the flip-flops.
- > The directed lines are labeled with two binary numbers separated by a slash.
- > The input value during the present state is labeled first, and the number after the slash gives the output during the present state with the given input.
- > For example, the directed line from state 00 to 01 is labeled 1/0, meaning that when the sequential circuit is in the present state 00 and the input is 1, the output is 0.
- > After the next clock cycle, the circuit goes to the next state, 01.

5. A sequential circuit has two D flip-flop, one input x , and one output y . Derive the state table and state diagram of the sequential circuit.



SOLUTION:

1. State equation

$$A(t+1) = Ax + Bx$$

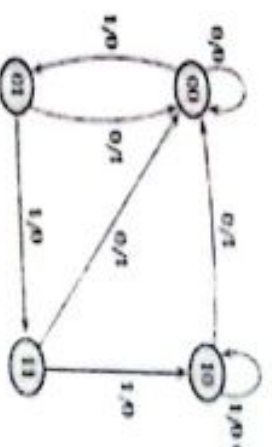
$$B(t+1) = Ax'$$

$$y = Ax' + Bx'$$

2. State table

Present State		Input		Next State		Output
A	B	x	x'	A	B	y
0	0	0	1	0	0	0
0	0	1	0	0	1	0
0	1	0	1	0	1	1
0	1	1	0	1	1	0
1	0	0	1	0	0	1
1	0	1	0	1	0	0
1	1	0	1	1	1	1
1	1	1	0	0	1	0

3. State diagram

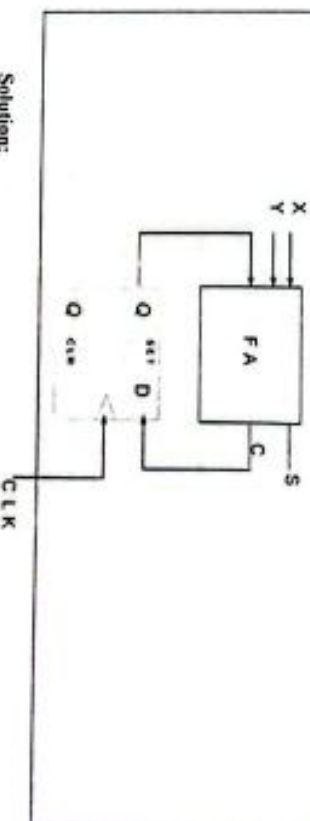


6. Construct a JK flip-flop using a D flip-flop, a 4-to-1-line multiplexer and an inverter.

Solution:



7. A sequential circuit has one flip-flop Q, two inputs x and y, and one output S. It consists of a full-adder circuit connected to a D flip-flop, as shown. Derive the state table and state diagram of the sequential circuit.



Solution:

1. FA equations: $S = x \oplus y \oplus Q$

2. Input equations: $C = xy + xQ + yQ$

$$D_0 = C$$

$$= xy + xQ + yQ \text{ (from the FA equations or from the K-map)}$$

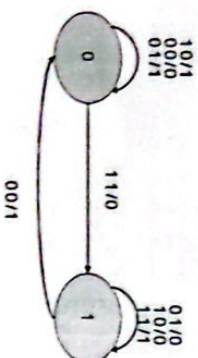
3. Characteristic equation:
 $Q(t+1) = D = XY + XQ + YQ$

4. State equation:
 $Q(t+1) = C$

5. State Table:

PRESENT STATE	INPUTS		NEXT STATE	OUTPUT
Q	X	Y	Q	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	1	0
1	0	1	1	0
1	1	0	0	1
1	1	1	1	0

State Diagram:



8. A sequential circuit has two JK flip-flops A and B and one input x. The circuit is described by the following flip-flop input equations:

$$J_A = X \quad K_A = B'$$

$$J_B = X \quad K_B = A$$

- Derive the state equation $A(t+1)$ and $B(t+1)$ by substituting the input equations for the J and K variables.
- Draw the state diagram of the circuit.

Solution:

1. State equation:

$$Q(t+1) = JQ' + KQ$$

2. Characteristic equation:

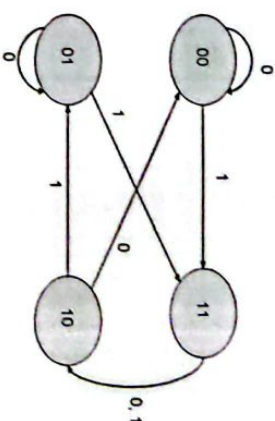
$$A(t+1) = XA' + BA$$

$$B(t+1) = XB' + AB$$

3. State Table:

PRESENT STATE	INPUT		NEXT STATE		FLIP-FLOP INPUTS			
A	B	X	A	B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	1	0	0
0	0	1	1	1	1	1	1	0
0	1	0	0	1	0	0	0	0
0	1	1	1	1	1	0	1	0
1	0	0	0	0	0	1	0	1
1	0	1	0	1	1	1	1	1
1	1	0	1	0	0	0	0	1
1	1	1	1	0	1	0	1	1

4. State Diagram:



9. Explain in detail about design of clocked sequential circuits.

The procedure for designing synchronous sequential circuits can be summarized by a list of recommended steps:

- From the word description and specifications of the desired operation, derive a state diagram for the circuit.
- Reduce the number of states if necessary.
- Assign binary values to the states.
- Obtain the binary-coded state table.
- Choose the type of flip-flops to be used.

6. Derive the simplified flip-flop input equations and output equations.
7. Draw the logic diagram.

10. Reduce the number of states in the following table and tabulate the reduced state table.

PRESENT STATE	NEXT STATE		OUTPUT	
	X=0	X=1	X=0	X=1
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	e	a	1	0
e	d	c	0	0
f	f	b	1	1
g	e	b	0	1
h	e	a	1	0

Solution:

States b,e are the same, we will replace state e with state b.
States d,h are the same, we will replace state h with state d.

PRESENT STATE	NEXT STATE		OUTPUT	
	X=0	X=1	X=0	X=1
a	f	b	0	0
b	d	c	0	0
c	f	b	0	0
d	e	a	1	0
f	f	b	1	1
g	e	d	0	1

States a,c are the same, we will replace state c with state a.

PRESENT STATE	NEXT STATE		OUTPUT	
	X=0	X=1	X=0	X=1
a	f	b	0	0
b	d	a	0	0
d	e	a	1	0
f	f	b	1	1
g	e	d	0	1

11. Starting from state a, and the input sequence 01110010011, determine the output sequence for:
(a) the state table of the previous problem and

- (b) the reduced state table from the previous problem. Show that the same output sequence is obtained for both.

Solution:

(a) using the state table

state	a	f	b	c	d	e	g	h	g	h	a
input	0	1	1	1	0	0	1	0	1	1	
output	0	1	0	0	0	1	1	0	1	0	

(b) Using the reduced state table

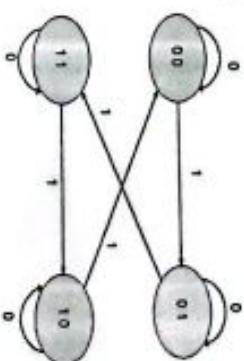
state	a	f	b	a	b	d	e	d	e	d	a
input	0	1	1	1	0	0	1	0	0	1	1
output	0	1	0	0	0	1	1	1	0	1	0

The same output sequence is obtained for both.

12. Design a sequential circuit with two D flip-flops A and B, and one input x. When x=0, the state of the circuit remains the same. When x=1, the circuit goes through the state transitions from 00 to 01 to 10 back to 00, and repeats.

Solution:

1. State Diagram:



2. State Table:

PRESENT STATE	INPUT		NEXT STATE	
	A	B	A	B
0	0	0	0	0
0	0	0	1	0
0	1	0	0	1
0	1	0	0	1
1	0	1	1	1
1	0	1	0	0
1	1	0	1	0
1	1	0	1	1
1	1	1	1	0

3. Characteristic equation:

$$Q(t+1) = D$$

4. Input equations or State equations:

$$A(t+1) = D_A(A,B,X) = \sum(3,4,6,7) \quad B(t+1) = D_B(A,B,X) = \sum(1,2,3,6)$$

K-maps:

D_A

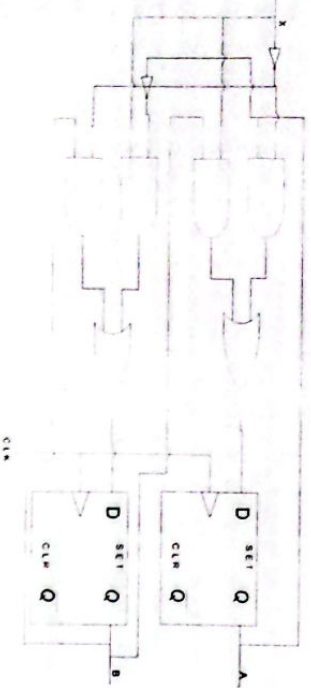
	A	BX			
A	0	00	01	11	10
0	0			1	
1	1				1

D_B

	A	BX			
A	0		11	10	
0			1		
1				1	

$$D_B = AX + BX'$$

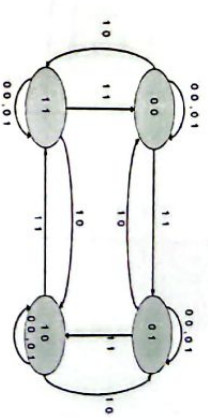
Circuit Diagram:



10 to 01 back to 00, and repeats.

Solution:

1. State Diagram:



2. State Table:

PRESENT STATE		INPUT		NEXT STATE		FLIP-FLOP INPUTS			
A	B	E	X	A	B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	0	X	0	X
0	0	0	1	0	0	0	X	0	X
0	0	1	0	1	1	1	X	1	X
0	0	1	1	1	1	1	X	1	X
0	1	0	0	1	0	1	0	X	0
0	1	0	1	1	0	1	0	X	0
0	1	1	0	0	0	0	X	X	1
0	1	1	1	0	0	0	X	X	1
1	0	0	0	1	0	1	0	0	X
1	0	0	1	1	0	1	0	0	X
1	0	1	0	1	1	1	X	1	X
1	0	1	1	1	1	1	X	1	X
1	1	0	0	1	1	1	X	0	0
1	1	0	1	1	1	1	X	0	0
1	1	1	0	0	0	0	X	0	1
1	1	1	1	0	0	0	X	0	1

K-maps:

J_A

	AB	00	01	11	10
0	0	0	0	0	1
1	0	0	0	1	0
11	X	X	X	X	X
10	X	X	X	X	X

$$J_A = BEX + B'EX' = E(B \oplus X)'$$

K_A

	AB	00	01	11	10
0	0	X	X	X	X
1	0	X	X	X	X
11	X	X	X	X	X
10	X	X	X	X	X

$$K_A = BEX + BE'X' = E \oplus B \oplus XY$$

$$J_B = E$$

J_B	AB	00	01	11	10
0	0	0	0	1	1
1	0	X	X	X	X
1	1	X	X	X	X
0	1	0	0	1	1

$$J_B = E$$

$$K_B = E$$

K_B	AB	00	01	11	10
0	0	X	X	X	X
1	0	0	0	1	1
1	1	0	0	1	1
0	1	X	X	X	X

Circuit Diagram:



This sequential circuit behaves like a 2-bit up-down-counter, with E: the enable of the whole counter, and resets when it finishes counting, when $X=1$, it behaves like an up-counter, when $X=0$, it behaves like a down-counter

14. Describe in detail about ripple counter/asynchronous counter?

- > Counters come in two categories: ripple counters and synchronous counters.
- > In a ripple counter, the flip-flop output transition serves as a source for triggering other flip-flops. In other words, the CP inputs of all flip-flops (except the first) are triggered not by the incoming pulses, but rather by the transition that occurs in other flip-flops.
- > In a synchronous counter, the input pulses are applied to all CP inputs of all flip-flops.

- > The change of state of a particular flip-flop is dependent on the present state of other flip-flops.
- > Some common MSI ripple counters are Binary Ripple Counter and BCD Ripple Counter.

1. Binary Ripple Counter

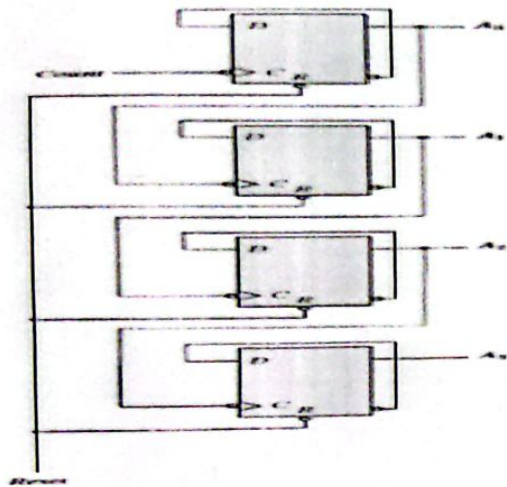
- > A binary ripple counter consists of a series connection of complementing flip-flops (T or JK type), with the output of each flip-flop connected to the CP input of the next higher-order flip-flop.
- > The flip-flop holding the least significant bit receives the incoming count pulses.
- > The small circle in the CP input indicates that the flip-flop complements during a negative-going transition or when the output to which it is connected goes from 1 to 0.
- To understand the operation of the binary counter, refer to its count sequence given in Table 1.
- It is obvious that the lowest-order bit A_1 must be complemented with each count pulse.
- Every time A_1 goes from 1 to 0, it complements A_2 .
- Every time A_2 goes from 1 to 0, it complements A_3 , and so on.

Table 1: Count Sequence for a Binary Ripple Counter

Count Sequence				Comments for Complementing Flip-Flops			
A_3	A_2	A_1	A_0	Complement A_0	Complement A_1	Complement A_2	Complement A_3
0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0
0	0	1	0	0	1	0	0
0	0	1	1	0	0	1	0
0	1	0	0	1	1	0	0
0	1	0	1	0	0	1	0
0	1	1	0	1	1	0	0
0	1	1	1	0	0	1	0
1	0	0	0	1	1	0	1
1	0	0	1	0	0	1	1
1	0	1	0	1	1	0	1
1	0	1	1	0	0	1	1
1	1	0	0	1	1	0	1
1	1	0	1	0	0	1	1
1	1	1	0	1	1	0	1
1	1	1	1	0	0	1	1

and so on

The below diagram shows the 4-bit binary ripple counter.



2. BCD Ripple Counter

A decimal counter follows a sequence of ten states and returns to 0 after the count of 9. Such a counter must have at least four flip-flops to represent each decimal digit, since a decimal digit is represented by a binary code with at least four bits. The sequence of states in a decimal counter is dictated by the binary code used to represent a decimal digit. If BCD is used, the sequence of states is as shown in the state diagram of Fig. 2.

Signals that affect the flip-flop transition depend on the order in which they change from 1 to 0. The operation of the counter can be explained by a list of conditions for flip-flop transitions. These conditions are derived from the logic diagram and from knowledge of how a JK flip-flop operates. Remember that when the CP input goes from 1 to 0, the flip-flop is set if $J = 1$, is cleared if $K = 1$, is complemented if $J = K = 1$, and is left unchanged if $J = K = 0$. The following are the conditions for each flip-flop state transition:

1. Q1 is complemented on the negative edge of every count pulse.
2. Q2 is complemented if Q8 = 0 and Q1 goes from 1 to 0. Q2 is cleared if Q8 = 1 and Q1 goes from 1 to 0.
3. Q4 is complemented when Q2 goes from 1 to 0.

4. Q8 is complemented when Q4 Q2 = 11 and Q1 goes from 1 to 0. Q8 is cleared if either Q4 or Q2 is 0 and Q1 goes from 1 to 0.

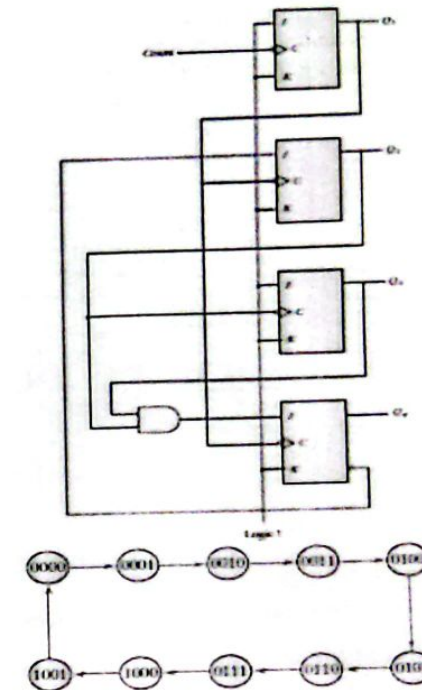


Figure: BCD ripple counter

15. Describe in detail about synchronous counter?

Synchronous counters are distinguished from ripple counters in that clock pulses are applied to the CP inputs of all flip-flops. The common pulse triggers all the flip-flops simultaneously, rather than one at a time in succession as in a ripple counter. If $J = K = 0$, the flip-flop remains unchanged. If $J = K = 1$, the flip-flop complements.

1. Synchronous binary counter

In a synchronous binary counter, the flip-flop in the lowest-order position is complemented with every pulse. This means that its J and K inputs must be maintained at logic-1. A flip-flop in any other position is complemented with a pulse provided all the bits in the lower-

order positions are equal to 1, because the lower-order bits (when all 1's) will change to 0's on the next count pulse. The binary count dictates that the next higher-order bit be complemented.

For example, if the present state of a 4-bit counter is $A_4A_3A_2A_1 = 0011$, the next count will be 0100. A_1 is always complemented. A_2 is complemented because the present state of $A_1 = 1$. A_3 is complemented because the present state of $A_2A_1 = 11$. But A_4 is not complemented because the present state of $A_3A_2A_1 = 011$, which does not give an all-1's condition.

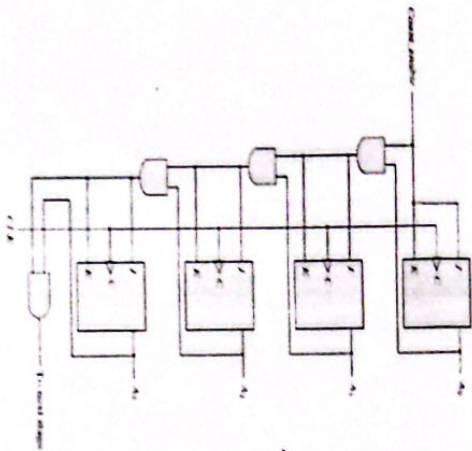


Figure: synchronous binary counter

2. Binary Up-Down Counter

A count-down binary counter can be constructed as shown in Fig. , except that the inputs to the AND gates must come from the complement outputs Q' and not from the normal outputs Q of the previous flip-flops. The two operations can be combined in one circuit.

- When the up input control is 1, the circuit counts up, since the T inputs receive their signals from the values of the previous normal outputs of the flip-flops.
- When the down input control is 1 and the up input is 0, the circuit counts down, since the complemented outputs of the previous flip-flops are applied to the T inputs.
- When the up and down inputs are both 0, the circuit does not change state but remains in the same count.
- When the up and down inputs are both 1, the circuit counts up. This ensures that only one operation is performed at any given time.

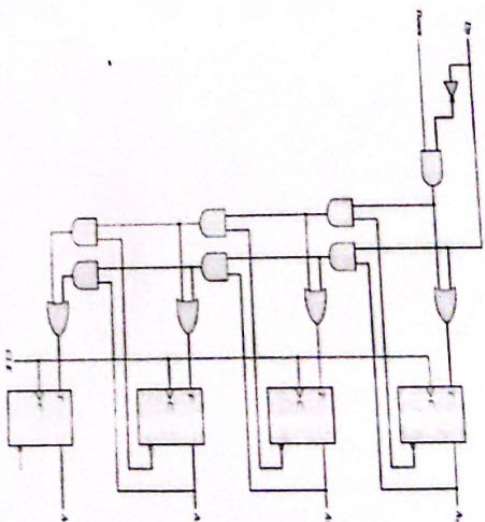


Figure: binary up-down counter

16. Elaborate decade counter.

To verify that these conditions result in the sequence required by a BCD ripple counter, it is necessary to verify that the flip-flop transitions indeed follow a sequence of states as specified by the state diagram.

- Q_1 changes state after each clock pulse.
- Q_2 complements every time Q_1 goes from 1 to 0, as long as $Q_8 = 0$.
- When Q_8 becomes 1, Q_2 remains at 0.
- Q_4 complements every time Q_2 goes from 1 to 0.
- Q_8 remains at 0 as long as Q_2 or Q_4 is 0.
- When both Q_2 and Q_4 become 1, Q_8 complements when Q_1 goes from 1 to 0.
- Q_8 is cleared on the next transition of Q_1 .
- The BCD counter of Fig. 1 is a decade counter, since it counts from 0 to 9.

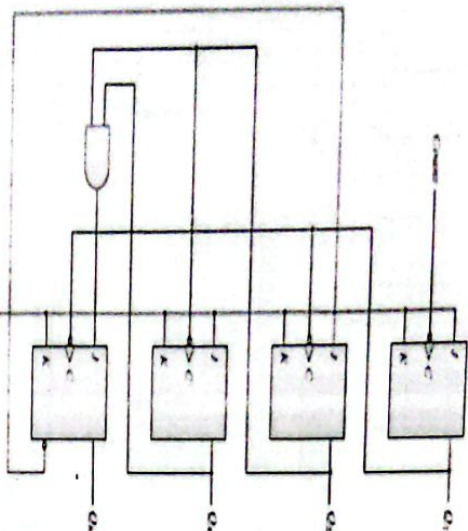


Figure 1: BCD counter

- > To count in decimal from 0 to 99, we need a two-decade counter.
- > To count from 0 to 999, we need a three-decade counter.
- > Multiple decade counters can be constructed by connecting BCD counters in cascade, one for each decade.
- > A three-decade counter is shown in Fig.2.
- > The inputs to the second and third decades come from Q_8 of the previous decade.
- > When Q_8 in one decade goes from 1 to 0, it triggers the count for the next higher order decade while its own decade goes from 9 to 0.

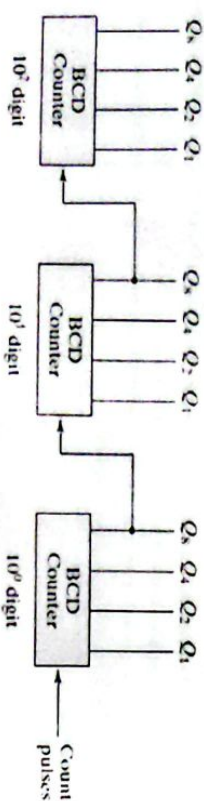
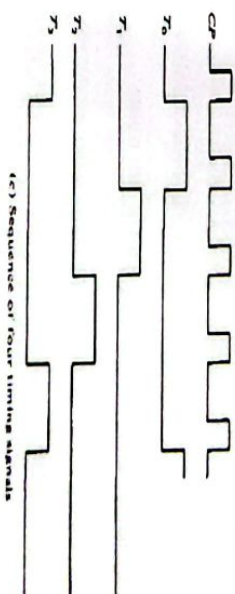
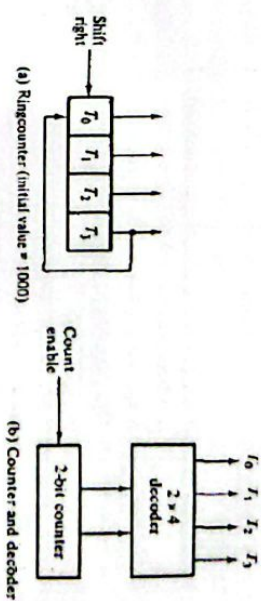


Figure 2: three decade decimal BCD counter

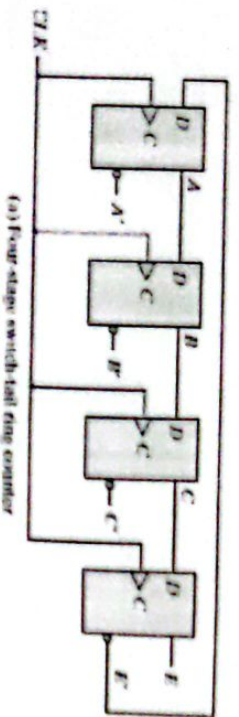
17. Explain Johnson counter and ring counter.

Ring counter

A *ring counter* is a circular shift register with only one flip-flop being set at any particular time, all others are cleared. The single bit is shifted from one flip-flop to the other to produce the sequence of timing signals. The below figure shows a 4-bit shift register connected as a ring counter. The initial value of the register is 1000, which produces the variable T_0 . The single bit is shifted right with every clock pulse and circulates back from T_3 to T_0 .

Johnson counter

- A k-bit ring counter circulates a single bit among the flip-flops to provide k distinguishable states.
- The number of states can be doubled if the shift register is connected as a *switch-tail* ring counter.
- A switch-tail ring counter is a circular shift register with the complement output of the last flip-flop connected to the input of the first flip-flop.
- In general, a k-bit switch-tail ring counter will go through a sequence of $2k$ states.
- Starting from all 0's, each shift operation inserts 1's from the left until the register is filled with all 1's.
- In the following sequences, 0's are inserted from the left until the register is again filled with all 0's.



(a) Four-stage switch tail ring counter

Sequence number	Flip-flop outputs				A to D gates required for sequence
	A	B	C	D	
1	0	0	0	0	$A \cdot B \cdot C \cdot D$
2	0	0	0	1	$A \cdot B \cdot C \cdot D'$
3	0	0	1	0	$A \cdot B \cdot C' \cdot D$
4	0	0	1	1	$A \cdot B \cdot C' \cdot D'$
5	0	1	0	0	$A \cdot B' \cdot C \cdot D$
6	0	1	0	1	$A \cdot B' \cdot C \cdot D'$
7	0	1	1	0	$A \cdot B' \cdot C' \cdot D$
8	0	1	1	1	$A \cdot B' \cdot C' \cdot D'$

(b) A constant sequence may need redundant decoders

18. What is register? Explain the types of shift register.

Register

Register is a group of flip flops for storing binary information

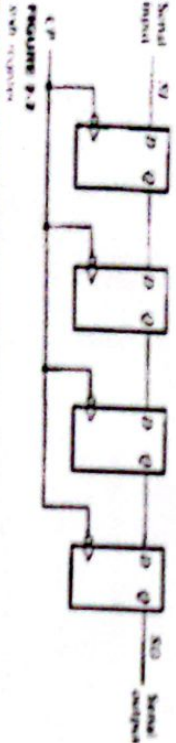
Shift register

A register capable of shifting its binary information either to the right or to the left is called a shift register.

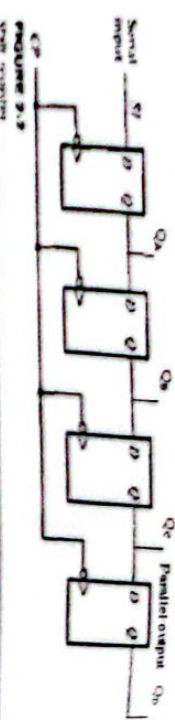
Types of shift register

1. Serial in serial out shift register
2. Serial in parallel out
3. Parallel in serial out
4. Parallel in parallel out

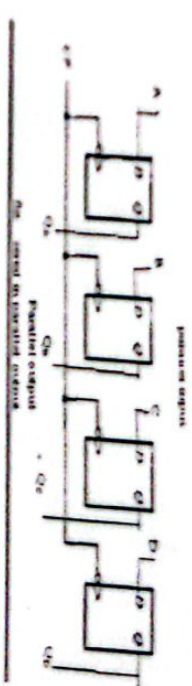
Serial in serial out shift register



Serial in parallel out shift register



parallel in parallel out shift register



Universal shift register

Shift registers provide the necessary input and output terminals for parallel transfer. They may also have both shift right and shift left capabilities. The most general shift register has the following capabilities:

Functions of Universal Shift Register

1. A clear control to clear the register to 0
2. A CP input for clock pulses to synchronize all operations.
3. A shift-right control to enable the shift-right operation and the serial input and output lines associated with the shift right
4. A shift-left control to enable the shift-left operation and the serial input and output lines associated with the shift left
5. A parallel-load control to enable a parallel transfer and the n input lines associated with the parallel transfer parallel output lines.
7. A control state that leaves the information in the register unchanged even though clock pulses are continuously applied

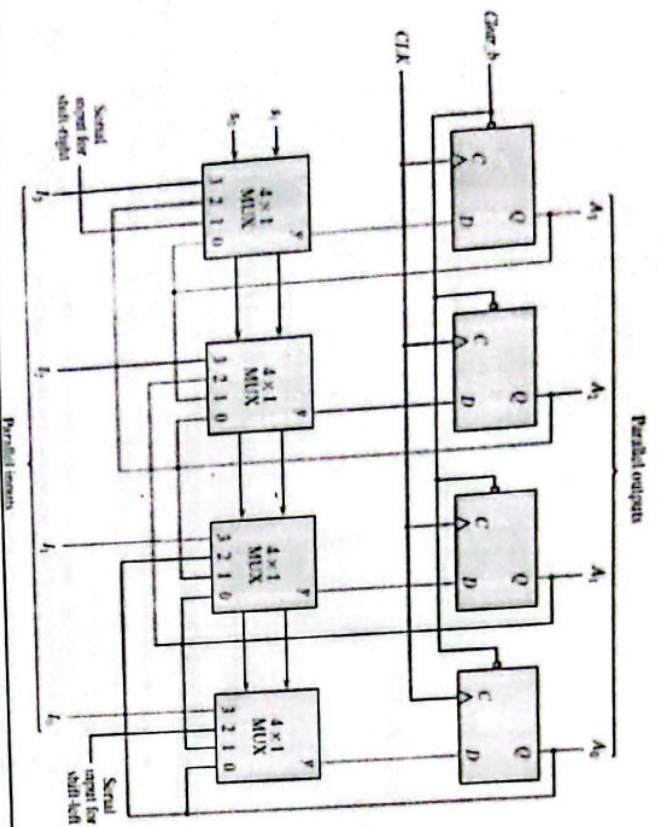


Figure: Four-bit universal shift register

Function Table for the Register of Fig. 6.7

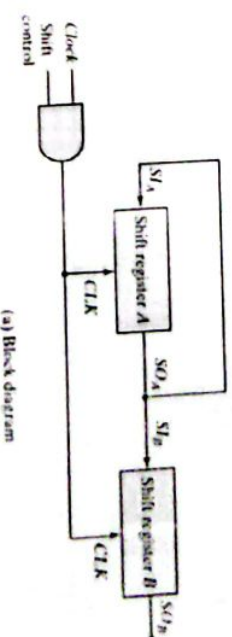
Mode Control		Register Operation
s1	s0	
0	0	No change
0	1	Shift right
1	0	Shift left
1	1	Parallel load

- > The circuit consists of four D flip-flops and four multiplexers.
- > The four multiplexers have two common selection inputs s1 and s0.

- > Input 0 in each multiplexer is selected when $s1s0 = 00$, input 1 is selected when $s1s0 = 01$, and similarly for the other two inputs. The selection inputs control the mode of operation of the register according to the function entries in Table.
- > When $s1s0 = 00$, the present value of the register is applied to the D inputs of the flip-flops.
- > This condition forms a path from the output of each flip-flop into the input of the same flip-flop, so that the output recirculates to the input in this mode of operation.
- > The next clock edge transfers into each flip-flop the binary value it held previously, and no change of state occurs.
- > When $s1s0 = 01$, terminal 1 of the multiplexer inputs has a path to the D inputs of the flip-flops. This causes a shift-right operation, with the serial input transferred into flip-flop A3.
- > When $s1s0 = 10$, a shift-left operation results, with the other serial input going into flip-flop A0. Finally, when $s1s0 = 11$, the binary information on the parallel input lines is transferred into the register simultaneously during the next clock edge.
- > Note that data enters MSB_{in} for a shift-right operation and enters LSB_{in} for a shift-left operation.
- > $Clear_b$ is an active-low signal that clears all of the flip-flops.

19. Explain in detail about Serial Addition using shift register.

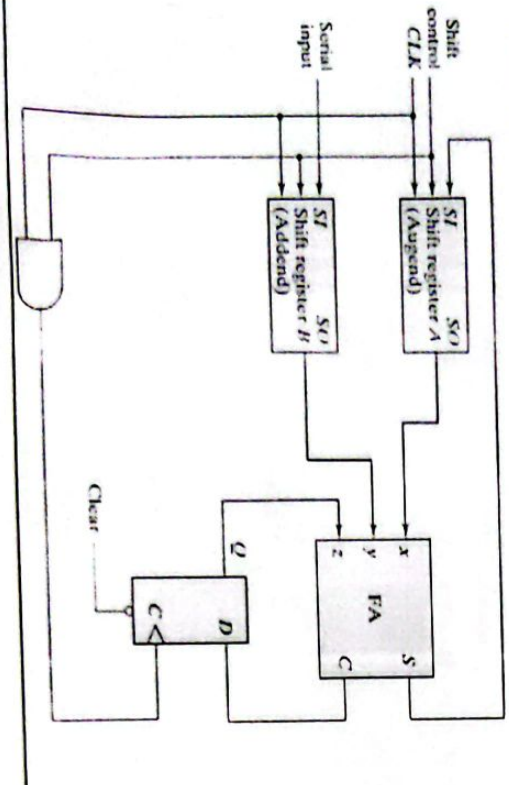
- > The serial transfer of information from register A to register B is done with shift registers, as shown in the block diagram of Figure.



(a) Block diagram

Serial Transfer Example	
Timing Pulse	Shift Register A Shift Register B
Initial value	1 0 1 1 0 0 1 0
After T ₁	1 1 0 1 1 0 0 1
After T ₂	1 1 1 0 1 1 0 0
After T ₃	0 1 1 1 0 1 1 0
After T ₄	1 0 1 1 1 0 1 1

- > The serial output (SO) of register A is connected to the serial input (SI) of register B.
- > To prevent the loss of information stored in the source register, the information in register A is made to circulate by connecting the serial output to its serial input.
- > The two binary numbers to be added serially are stored in two shift registers.
- > The operation of the serial adder is as follows: Initially, register A holds the augend, register B holds the addend, and the carry flip-flop is cleared to 0.
- > The outputs (SO) of A and B provide a pair of significant bits for the full adder at x and y.
- > Output Q of the flip-flop provides the input carry at z.
- > Initially, register A and the carry flip-flop are cleared to 0, and then the first number is added from B. While B is shifted through the full adder, a second number is transferred to it through its serial input.
- > The second number is then added to the contents of register A, while a third number is transferred serially into register B.
- > This can be repeated to perform the addition of two, three, or more four-bit numbers and accumulate their sum in register A.



State Table for Serial Adder

Present State	Inputs	Next State	Output	Flip-flop Inputs
Q	x y	Q	S	J _Q K _Q
0	0 0	0	0	0 X
0	0 1	0	1	0 X
0	1 0	0	1	0 X
0	1 1	1	0	1 X
1	0 0	1	0	X 0
1	0 1	1	1	X 0
1	1 0	1	1	X 0
1	1 1	0	0	X 1

State equation

$$J_Q = xy$$

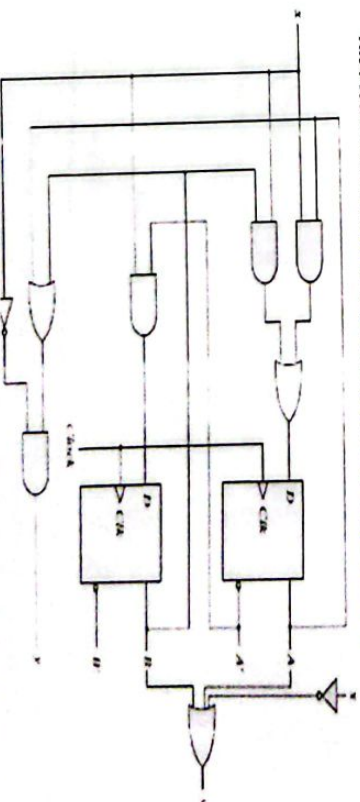
$$K_Q = x'y' = (x + y)'$$

$$S = x \oplus y \oplus Q$$

20. Explain in detail Mealy and Moore Models of Finite State Machines.

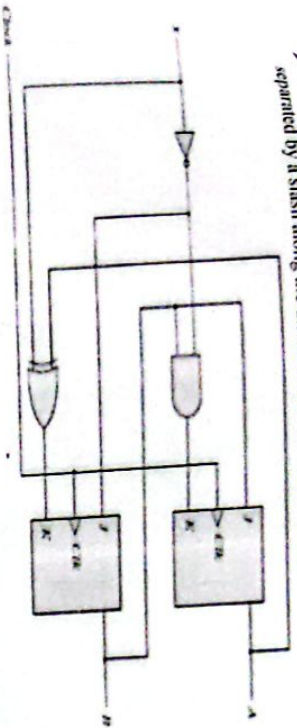
Mealy and Moore Models of Finite State Machines

- > The most general model of a sequential circuit has inputs, outputs, and internal states.
- > It is customary to distinguish between two models of sequential circuits: the Mealy model and the Moore model.
- > They differ only in the way the output is generated.
- > In the Mealy model, the output is a function of both the present state and the input.
- > In the Moore model, the output is a function of only the present state.
- > A circuit may have both types of outputs. The two models of a sequential circuit are commonly referred to as a finite state machine, abbreviated FSM.
- > The Mealy model of a sequential circuit is referred to as a Mealy FSM or Mealy machine. The Moore model is referred to as a Moore FSM or Moore machine.



... is an example of a Mealy machine.

- The circuit presented above is an example of a *combinational* circuit. The output y is a function of both input x and the present state of A and B . The corresponding state diagram in Fig. 5.16 shows both the input and output values. The corresponding state diagram in Fig. 5.16 shows both the input and output values, separated by a slash along the directed lines between the states.



- An example of a Moore model is given above.
- Here, the output is a function of the present state only.
- The output depends only on flip-flop values, and that makes it a function of the present state only.
- The input value in the state diagram is labeled along the directed line, but the output value is indicated inside the circle together with the present state.

In a Moore model, the outputs of the sequential circuit are synchronized with the clock, because they depend only on flip-flop outputs that are synchronized with the clock.

- In a Mealy model, the outputs may change if the inputs change during the clock cycle.
- Moreover, the outputs may have momentary false values because of the delay encountered from the time that the inputs change and the time that the flip-flop outputs change.
- In order to synchronize a Mealy-type circuit, the inputs of the sequential circuit must be synchronized with the clock and the outputs must be sampled immediately before the clock edge. The inputs are changed at the inactive edge of the clock to ensure that the inputs to the flip-flops stabilize before the active edge of the clock occurs.
- Thus, the output of the Mealy machine is the value that is present immediately before the active edge of the clock.

21. Synthesis of JK flip-flop

From Table and Pencil to Laptop

Present State		Next State		Flip-Flop Inputs			
A	B	A	B	J _A	K _A	J _B	K _B
0	0	0	0	0	0	X	0
0	0	1	0	1	0	X	1
0	1	0	1	0	1	X	0
0	1	1	1	1	0	X	0
1	0	0	1	0	1	X	0
1	0	1	1	1	0	X	0
1	1	0	0	0	1	X	0
1	1	1	0	1	0	X	1

